



SQL CLASS NOTE – MODULE 8: BETWEEN

1. Introduction

The **BETWEEN operator** is used to filter results **within a given range**.

- Works with **numbers, dates, and text**.
- By default, **BETWEEN** is **inclusive** → it includes the start and end values.

👉 Example:

- `WHERE Age BETWEEN 15 AND 18` → gives ages 15, 16, 17, and 18.

2. Syntax

```
SELECT column1, column2, ...  
FROM table_name  
WHERE column_name BETWEEN value1 AND value2;
```

- `value1` = lower bound.
- `value2` = upper bound.

You can also use `NOT BETWEEN` to exclude ranges.

3. Practical Examples with `SchoolDB`

Example 1: Students aged between 15 and 18

```
SELECT FirstName, LastName, Age  
FROM Students  
WHERE Age BETWEEN 15 AND 18;
```

Example 2: Students not aged between 10 and 14

```
SELECT FirstName, LastName, Age
FROM Students
WHERE Age NOT BETWEEN 10 AND 14;
```

Example 3: Enrollments between two dates

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
WHERE EnrollmentDate BETWEEN '2025-01-10' AND '2025-01-20';
```

Example 4: Students in grades between Grade 9 and Grade 11

```
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE GradeLevel BETWEEN 'Grade 9' AND 'Grade 11';
```

📌 Works because grades are stored as **text** and follow alphabetical order.

Example 5: Courses with codes between 100 and 200

```
SELECT CourseName, CourseCode
FROM Courses
WHERE CourseCode BETWEEN 100 AND 200;
```

4. Why Use BETWEEN ?

- Simplifies queries with ranges.
- Easier than chaining multiple conditions with `>=` and `<=`.
- Works on **dates**, which is very useful for time-based queries.

5. Real-Life Use Cases

- **Schools:** Get students with ages between 13 and 17.
- **Banks:** Find transactions between ₦10,000 and ₦50,000.
- **Hospitals:** Show patients admitted between two dates.
- **E-commerce:** List products priced between 100*and*500.

6. Classwork (Practice)

👉 Try these queries on **SchoolDB**:

1. Display all students aged between 14 and 17.
2. Show all students not aged between 12 and 15.
3. Find enrollments that happened between **2025-01-12** and **2025-01-18**.
4. Show all courses with `CourseCode` between 200 and 400.
5. Display all students whose grade level is between **Grade 10** and **Grade 12**.
6. Show all students whose first names fall alphabetically between “**J**” and “**S**”.
7. Display all enrollments where `EnrollmentDate` is not between **2025-01-10** and **2025-01-15**.

7. Challenge Question

👉 Write a query that shows the `FullName`, `Age`, and `GradeLevel` of all **female students aged between 15 and 18**, who are also enrolled between **2025-01-12** and **2025-01-18**.

✅ That's the **expanded Module 8: BETWEEN**.

Perfect 👉 Let's solve the **Module 8 Classwork + Challenge** step by step.



Solutions to Module 8: BETWEEN

1. Display all students aged between 14 and 17.

```
SELECT FirstName, LastName, Age, GradeLevel
FROM Students
WHERE Age BETWEEN 14 AND 17;
```

2. Show all students not aged between 12 and 15.

```
SELECT FirstName, LastName, Age, GradeLevel
FROM Students
WHERE Age NOT BETWEEN 12 AND 15;
```

3. Find enrollments that happened between 2025-01-12 and 2025-01-18.

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
WHERE EnrollmentDate BETWEEN '2025-01-12' AND '2025-01-18';
```

4. Show all courses with CourseCode between 200 and 400.

```
SELECT CourseName, CourseCode, Teacher
FROM Courses
WHERE CourseCode BETWEEN 200 AND 400;
```

5. Display all students whose grade level is between Grade 10 and Grade 12.

```
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE GradeLevel BETWEEN 'Grade 10' AND 'Grade 12';
```

6. Show all students whose first names fall alphabetically between “J” and “S”.

```
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE FirstName BETWEEN 'J' AND 'S';
```

📌 This uses alphabetical ordering.

7. Display all enrollments where EnrollmentDate is not between 2025-01-10 and 2025-01-15.

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
WHERE EnrollmentDate NOT BETWEEN '2025-01-10' AND '2025-01-15';
```



Challenge Question

👉 Show the FullName, Age, and GradeLevel of all **female students aged between 15 and 18**, who are also enrolled between **2025-01-12** and **2025-01-18**.

```
SELECT CONCAT(s.FirstName, ' ', s.LastName) AS FullName,
       s.Age,
       s.GradeLevel
FROM Students AS s
JOIN Enrollments AS e ON s.StudentID = e.StudentID
WHERE s.Gender = 'Female'
      AND s.Age BETWEEN 15 AND 18
      AND e.EnrollmentDate BETWEEN '2025-01-12' AND '2025-01-18';
```

📌 This combines:

- BETWEEN for **age**
- BETWEEN for **enrollment date**
- A filter for **female students**

✅ That wraps up **Module 8: BETWEEN**.

👉 Do you want me to continue with **Module 9: IN** in the same detailed style (definitions, examples, classwork, and challenge)?